

LiftMetrics

Video analysis of bar path and bar velocity
for Olympic lifting and strength training

Snatch · Clean · Clean & jerk · power versions · jerk variants

Back squat · Front squat · Overhead squat · Deadlift · Bench press

A lift · A complex · A session

Presentation and ambassador proposal

Prepared for Teti Xanthoudaki

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1 Introduction

In Olympic lifting, the difference between a made lift and a miss often comes down to a few centimetres: a bar that swings forward in the second pull, a catch too far in front, a drop under the bar that takes too long. These faults are hard to see with the naked eye, in real time, at the speed of a snatch.

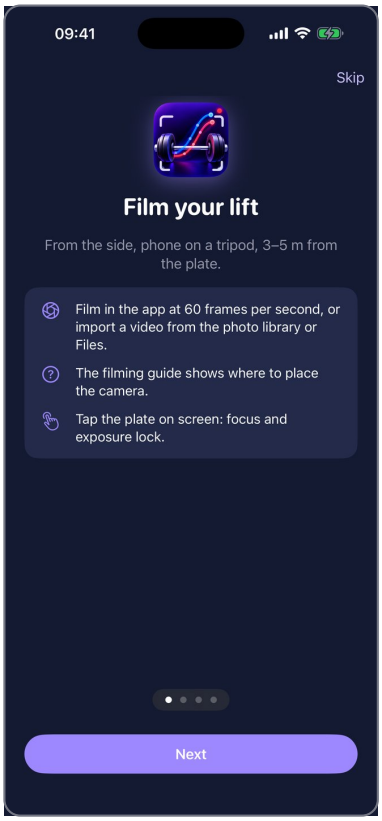
LiftMetrics is an iPhone app that **films the lift from the side, tracks the plate frame by frame and draws the bar path**. It splits the movement into phases, measures what a coach needs (horizontal displacements, heights, velocities, timings), gives a **technique score out of 100** and offers **concrete coaching**: cues and drills for each fault.

In short

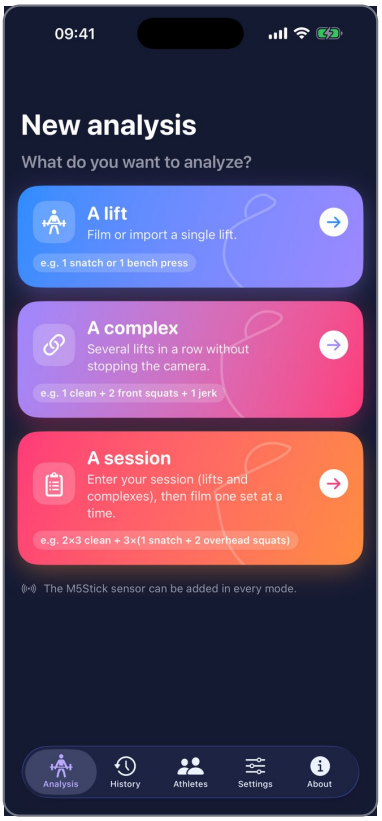
- An iPhone on a tripod is all you need: the side-view video is the only requirement.
- **Hands-free**: say “Go” to start filming and “Down” to stop — the app trims the video around the lift by itself.
- An optional M5StickS3 sensor on the bar sleeve makes velocity more accurate and measures bar tilt.
- Results in under a minute: bar path coloured by phase, curves, score, coaching cues.
- Squats (back, front, overhead), deadlift and bench press: film the whole set at once and get the velocity of every rep plus velocity loss (velocity-based training).
- A complex in **one continuous video**: the app splits it into lifts on its own.
- Comparison with the world’s best lifters (2018 IWF World Championships) and a national-level team.
- **Posture**: knee, hip, trunk and elbow angles at the key moments of the lift, drawn on the video frames.
- **Masters**: loads corrected for age (IMWA 2025 factors) and bodyweight (Sinclair).
- PDF reports, pictures and videos for social media, and a file format to share lifts between users.
- Everything is computed and stored on the iPhone: no account, no video uploaded.

Who is it for?

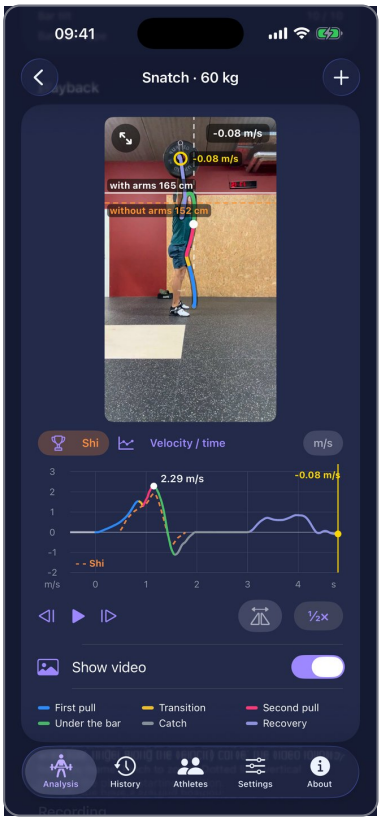
- **Athletes**, to understand their lifts and track progress session after session.
- **Coaches**, to back up what they see with numbers, compare an athlete’s lifts and build programmes.
- **Boxes, clubs and events**, to offer instant analysis to athletes and produce visuals to share.



Welcome screen (first launch)



Choosing the type of analysis



Replay of an analysed lift

2 Why it matters for CrossFit athletes

Barbell cycling, heavy singles and strength pieces are at the heart of CrossFit training and competition. **LiftMetrics** brings lab-style barbell feedback to any box, with nothing more than a phone:

- **Snatch and clean & jerk technique** under fatigue or at heavy loads: see exactly where the bar drifts, how deep and how fast the athlete drops under it, and whether the catch position holds.
- **Velocity-based strength work** on back squat, front squat, overhead squat, deadlift and bench press: mean and peak velocity of every rep and velocity loss across the set — a simple, objective way to manage fatigue during a training block.
- **Complexes and sessions**: chain several lifts in one take (for example 1 clean + 2 front squats + 1 jerk) or follow the day's programme set by set, with loads computed from the athlete's 1RMs.
- **Body positions**: knee, hip and trunk angles at lift-off, in the pulls, at the catch and at the bottom of the squat — with the jerk split checked front knee and back knee.
- **Training alone**: voice commands start and stop the camera, and the analysis can wait until the end of the set or the session — no need to touch the phone between lifts.
- **No extra hardware needed**: side-view video is enough. The sensor is optional, for those who want more precision.
- **Content ready to post**: a picture or slow-motion video of a lift with its bar path and live velocity, for any social network.

Typical use

The athlete says “Go”, snatches, says “Down” once the bar is held: the video is already trimmed. She taps the plate, sees the bar path and the score within a minute, then sends the lift file to her coach, who opens it in their own app with all the curves and measurements.

3 System and environment

Main device	iPhone running iOS 17 or later (iPhone app only, no iPad).
Camera	iPhone rear camera, 60 fps in HD 1080p (4K works but adds no useful precision).
Sensor (optional)	M5Stack **M5StickS3** (BMI270 accelerometer and gyroscope) running the “BarPathLogger” firmware; the M5StickC Plus2 is also supported.
Connections	Bluetooth for the sensor. No internet connection needed.
Voice commands	iPhone microphone; speech recognition runs on the iPhone (English words “Go”, “Down”, “Stop”), nothing is sent.
Languages	English, French, Spanish, German, Italian (chosen in Settings).
Data	Local database on the iPhone; videos in Photos or in Files → LiftMetrics. No server, no account.

Why the iPhone?

The iPhone camera natively films at 60 frames per second with a very steady frame rate, which makes velocity calculations reliable. All computations (plate tracking, phases, measurements) run on the device: analysis works in a gym with no network.

4 Required and optional hardware

Hardware	Status	Role
iPhone with LiftMetrics	**required**	Films (or imports) the side-view video, runs the analysis, keeps the history.
Tripod or stable stand	**required**	The camera must not move: any shake distorts the bar path.
Plates of known diameter	**required**	The plate facing the camera sets the scale (45 cm by default, adjustable).
M5StickS3 + sleeve mount	optional	More accurate velocity (200 samples/s), bar tilt and rotation, power.

4.1 Two possible setups

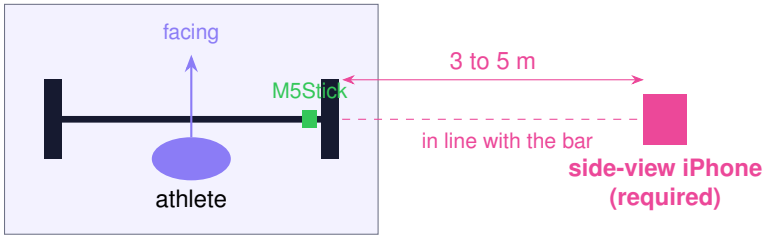
Setup	What you need	Best for
A — iPhone only	1 iPhone + tripod	Getting started, gyms with no equipment, importing existing videos.
B — iPhone + M5Stick	1 iPhone + tripod + sensor on the sleeve	Accurate tracking of velocity, power and bar tilt.

4.2 The M5StickS3 sensor

- Records the accelerometer (±16 g) and gyroscope (±2000 °/s) at **200 Hz** in its memory.
- Controlled by the app over Bluetooth: starts and stops with the camera, clock sync, file transfer then erase. Files from previous days are cleared automatically when it connects; today's are kept.
- Mounted on the sleeve **opposite the camera**: the app knows which side from the camera setting.
- Button A: start or stop recording (it can also trigger the iPhone: see §7).
- The screen shows whether it is recording on its own or on the iPhone's request.
- Fallback mode: Wi-Fi access point with a page to download the files.

5 Setup

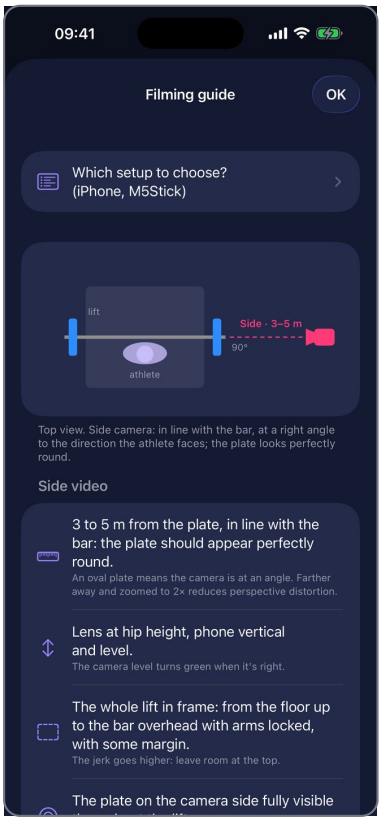
5.1 Positioning the camera



platform (top view)

The built-in filming guide recalls the key rules:

- Side view in line with the bar, 3–5 m away: the plate must look **perfectly round** (oval = camera at an angle). If the athlete is seen from the front or at three-quarters, the app warns that horizontal measurements are distorted by perspective.
- Camera on the athlete's left or right: set it once (lift, complex or session), or let the app detect which way the athlete faces.
- Lens at hip height, phone upright and level (the on-screen level turns green).
- The whole movement in frame, from the floor to arms locked overhead (the jerk goes higher).
- Plate facing the camera fully visible, nothing in front of it.
- 60 fps; tapping the plate on screen locks focus and exposure.
- Good light, avoid backlighting.
- Start 2 s before lift-off, stop 2 s after the catch: the still bar serves as a reference. With voice commands, "Go" and "Down" do this for you.

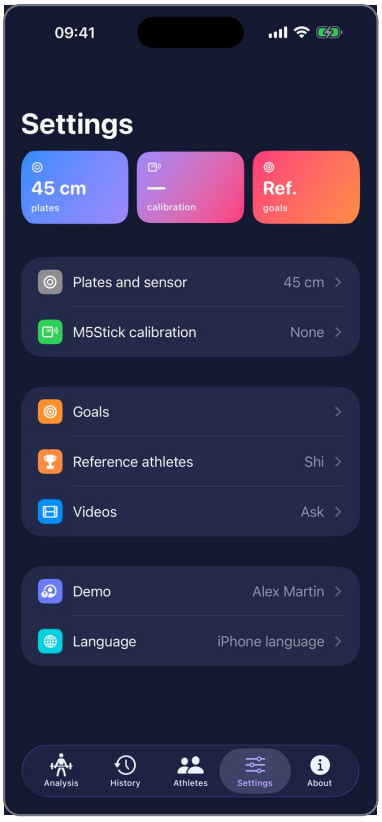


Filming guide

5.2 Settings

Settings are organised like on an iPhone, one page per topic:

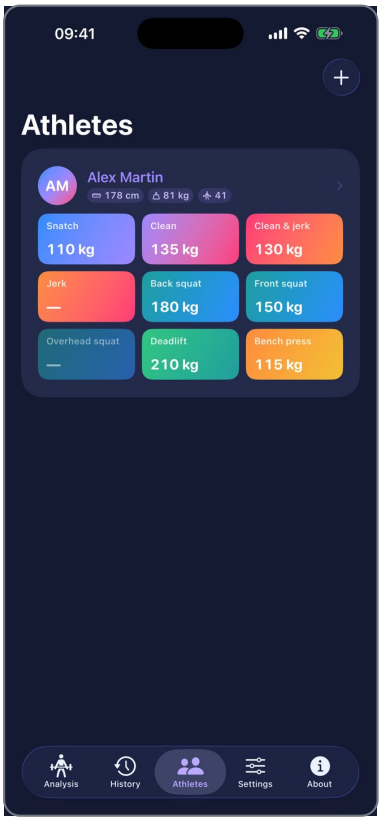
- **Plates and sensor**: plate diameter (scale), sleeve side where the sensor is mounted.
- **M5Stick calibration**: guided 6-position wizard (one per face), with audio cues.
- **Goals**: target ranges for each measurement, per movement (snatch, clean, jerk).
- **Reference athletes**: athlete shown on the charts and alignment mode (§17).
- **Voice commands**: on or off (see §8).
- **Videos**: after a lift is saved, delete full-length videos, ask, or keep everything (the app always keeps a trimmed clip of the lift).
- **Language**: the iPhone's language or one of the app's 5 languages.



Settings

5.3 Athletes

Each athlete has a profile: **height** (heights shown as % of height), **body mass** (power per kg), **category** women/men (automatic choice of the reference athlete), **date of birth** (Masters age) and **dated 1RMs** per movement. The percentage of max for a lift uses the 1RM valid on its date; a session programme computes loads from these 1RMs.



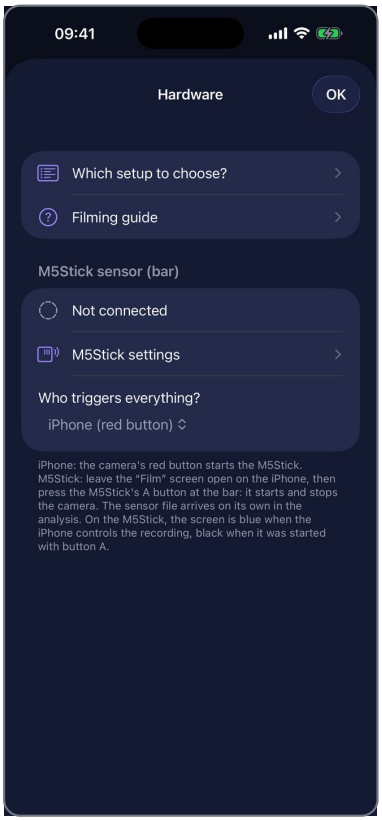
Athlete profile and 1RMs

5.4 Connected hardware

The **Hardware** screen connects the M5Stick, gives access to its settings and defines **who triggers everything**:

- **the iPhone**: the camera's red button starts the video and the sensor;
- **the M5Stick**: one press on the sensor's button A, mounted on the bar, starts and stops everything — handy when the athlete trains alone.

The sensor's battery level shows on the iPhone, and its file flows into the analysis automatically.



Hardware

6 Installation

1. **LiftMetrics** on the main iPhone: App Store (release planned) or TestFlight during the beta.
2. On first launch, a 4-page introduction explains the concept; it remains available in About.
3. **Permissions** requested when needed: camera, Photos (importing and saving videos), Bluetooth (sensor).
4. **Sensor option**: M5StickS3 with the BarPathLogger firmware (M5Unified library, "M5StickS3" board in the Arduino IDE). The sensor can be supplied pre-programmed. Connect it in Hardware, then run the guided calibration once.
5. Create the athlete profile (height, body mass, category, 1RMs).

Setup time

Setup A: 2 minutes (put the iPhone on a tripod). Setup B: 5 minutes the first time (pairing and calibration), then the app finds the hardware on its own.

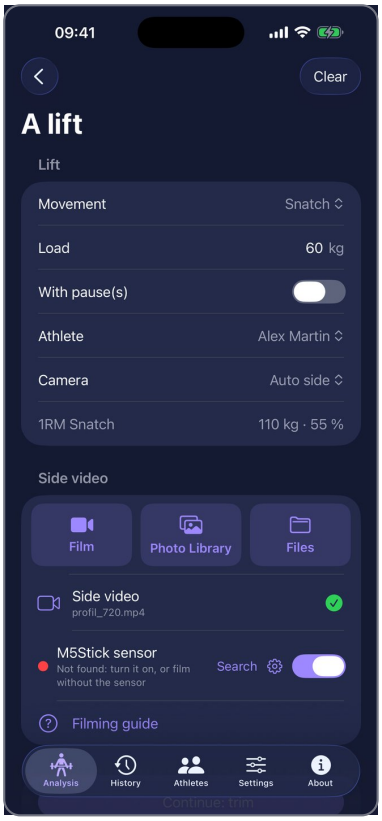
7 Modes

The home page of the **Analysis** tab offers three modes. Once a mode is chosen, the page only shows what is relevant to it; the "‹" arrow goes back to the choice.

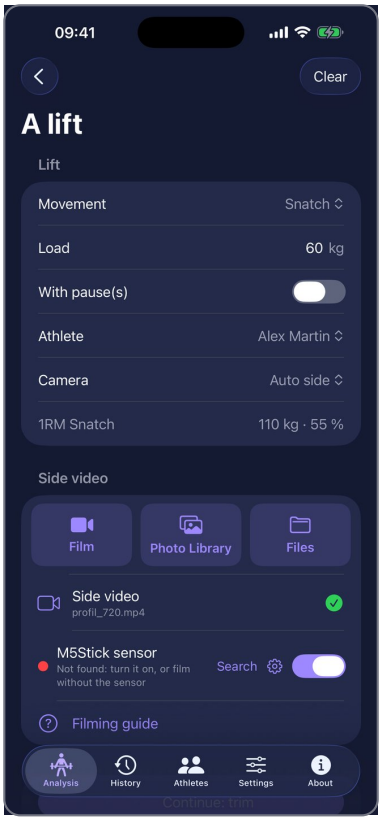
Mode	Principle	Example
A lift	One lift, filmed now or imported (Photos, Files).	1 snatch at 80 kg.
A complex	Several lifts chained without stopping the camera: one video, split into lifts automatically, analysed one by one and grouped in the history.	1 clean + 2 front squats + 1 jerk.
A session	The day's programme (lifts and complexes), guided set by set; loads computed from 1RMs.	2×3 cleans + 3×(1 snatch + 2 overhead squats).

7.1 A lift

Choose the movement (snatch, clean, clean & jerk, with or without pause; back squat, front squat, deadlift, bench press), the load and the athlete, then the side-view video: **film** with the connected devices, or **import** from Photos or Files (a sensor file can then be added).



A lift: movement, load, athlete, video



Connected devices and camera

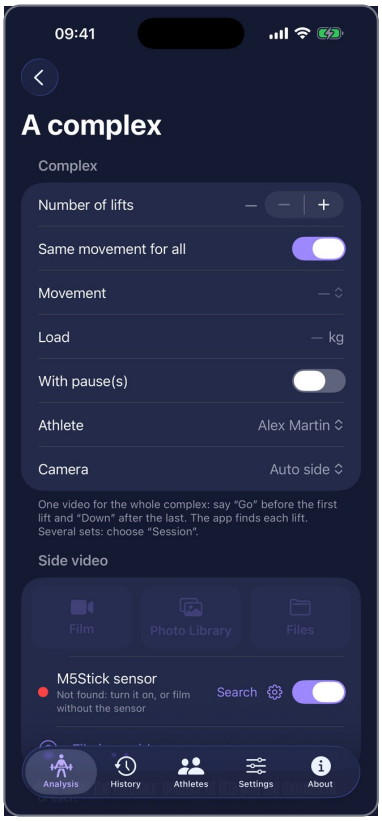
7.2 A complex

Number of sets, number of lifts per set and the movement of each lift (the same for all, or different). The whole complex is filmed **in one video**: the app finds each lift from the moments when the bar is still between them, and proposes a trim for each. If one trim is corrected by hand, the others stay automatic.

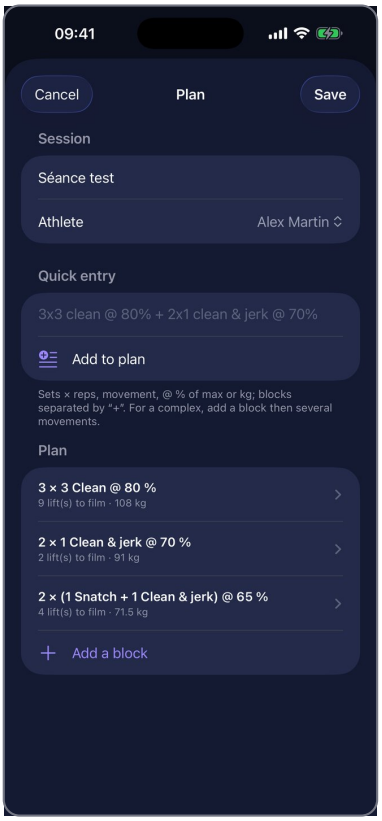
7.3 A session (programme)

The programme is typed in plain text (“2x3 clean + 3x(1 snatch + 2 overhead squat)”) or block by block; a block can be a complex. The app shows the next set and its computed load and lets you film it. Analysed lifts join the session in the history.

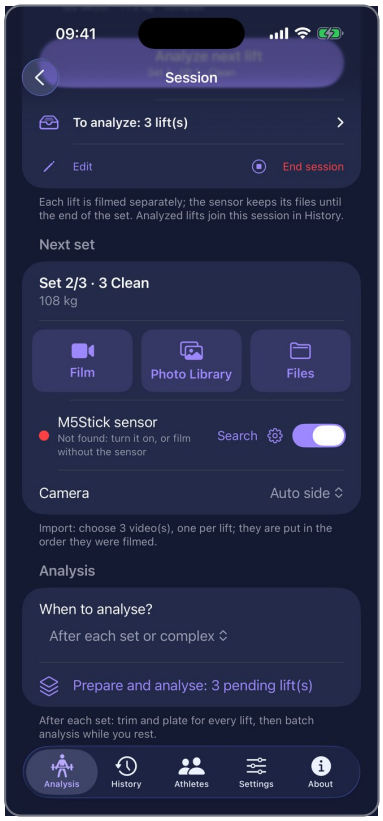
When to analyse? After each lift, after each set or complex, or at the end of the session. In the last two cases the lifts wait in a list: the athlete prepares each one in about ten seconds (trim and plate), then launches **one batch analysis** for all of them — during the rest between sets, or after training.



A complex



Preparing the programme



Session: analyse, then film the next set

8 Hands-free recording: voice commands

Training alone, the athlete does not need to walk to the phone. With **Voice commands** on (Settings), the camera listens for three English words:

Word	Signal	Effect
Go	1 beep	Starts the recording, just before the lift.
Down	2 beeps	Stops the recording 2.5 s later, once the bar is held (or back on the floor).
Stop	—	False start: the recording is cancelled, nothing is saved or transferred, and the set does not move on.

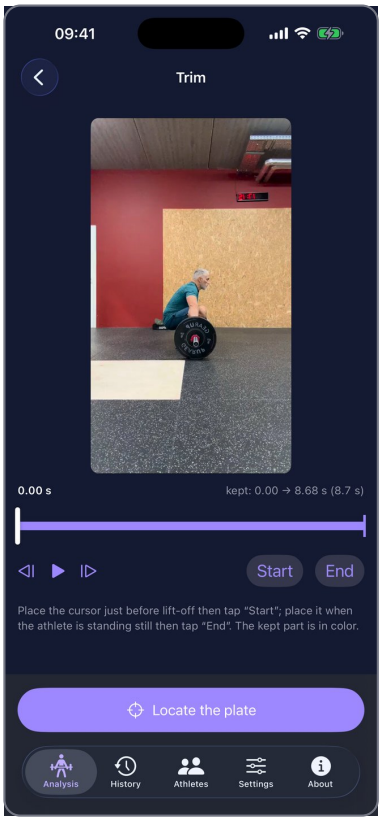
Automatic trim. After “Go”, the app looks for the moment the bar starts to move; after “Down”, for the moment it becomes still. The trim is placed around the lift and the next button reads **“Continue: locate the plate”**: the athlete goes straight to the plate. If the words or the movement are not clear enough, the app says why and falls back to manual trimming, pre-set between “Go” and “Down”.

Imported videos (Photos, Files) are listened to as well: a video filmed with another app, where the athlete said “Go” and “Down”, is trimmed the same way. Recognition runs entirely on the iPhone; the saved videos are silent.

9 Analysing a lift

9.1 Steps

1. **Video** filmed or imported (with the sensor file if any).
2. **Trim**: keep a few seconds around the lift — automatic with voice commands.
3. **Plate detection**: one tap on the plate; adjust the circle if needed (pinch to zoom).
4. **Analysis**: frame-by-frame plate tracking, scaling from the plate diameter, automatic sync with the sensor, phase detection, measurements.
5. **Results**: score, coaching, replay, measurements (under a minute on a recent iPhone).



Trimming around the lift

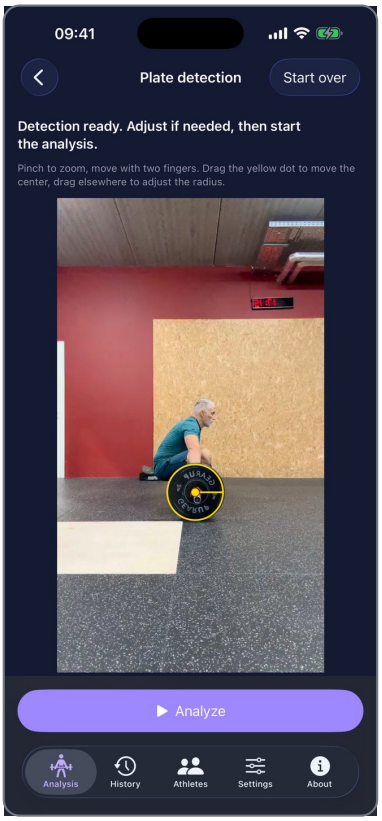
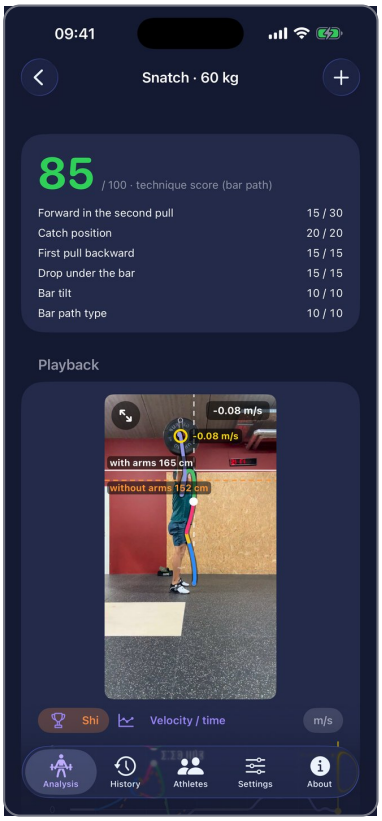


Plate detection



Score out of 100 and details

9.2 Phases of the movement

The bar path is coloured by phase, the same way on the video and on the curves:

Phase	Boundaries
First pull	from lift-off to the transition (end of knee extension)
Transition	knees re-bend under the bar
Second pull	from the transition to peak velocity
Under the bar	from peak velocity to the catch
Catch	stabilisation, then recovery to standing
Jerk	dip, drive, lockout

9.3 Checks and corrections

- **Lifter direction** from the camera-side setting, or detected automatically from the athlete's posture, editable.
- **Camera angle**: a warning when the athlete is not filmed side-on.
- **Tracking correction**: if the plate is briefly hidden, it can be repositioned and tracking restarted from that point.
- **Video–sensor sync** can be checked and adjusted to the hundredth of a second.
- **Re-analysis** of a lift from the history using its saved videos and files.

10 Measurements by hardware

All horizontal measurements are relative to the vertical line of the bar's starting position; heights are measured from the floor (plate centre) and as % of the athlete's height when known.

Measurement	iPhone only	+ M5Stick
Bar path		
Bar path coloured by phase	✓	✓
Bar sweep in the first pull (bar pulled back toward the athlete)	✓	✓
Forward displacement in the second pull	✓	✓
Loop and catch position	✓	✓
Bar path type (1, 2 or 3)	✓	✓
Peak height, catch height, drop under the bar	✓	✓
Height without arm pull, arm contribution	✓	●
Velocity and timing		
Peak velocity, velocity at the end of the first pull	✓	●
Mean velocity, first and second pull	✓	●
Peak acceleration, first and second pull	✓	●
Durations: first pull, second pull, turnover	✓	●
Catch stabilisation, recovery, pauses	✓	✓
Peak power (W and W/kg)	—	✓
Bar orientation		
Peak tilt (one side higher)	—	✓
Peak horizontal rotation (one side ahead of the other)	—	✓
Jerk		
Dip depth and duration	✓	●
Peak drive velocity	✓	●
Displacement during the drive, lockout position	✓	✓

✓ measured ● measured more accurately with the sensor (200 samples/s, orientation tracked by the gyroscope) — not available

Video velocity vs sensor velocity

Without a sensor, velocity is computed from the plate position 60 times per second. With the M5Stick, acceleration measured 200 times per second is fused with the video: the curve is finer, especially at peak velocity and during the turnover. The moments when the bar is still, before and after the lift, are used to re-anchor the velocity.

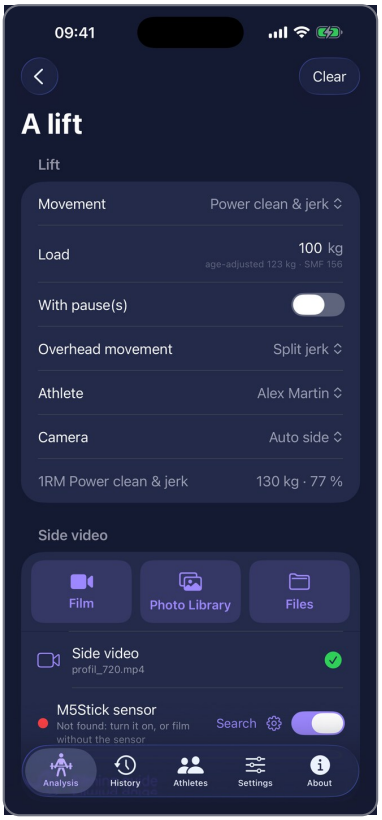
11 Power lifts and jerk variants

Power snatch, power clean and power clean & jerk are movements of their own in the list, next to the full lifts. They get the same bar path, measurements, score and coaching, but they are kept apart in the history, the charts and the review by movement, so power and squat lifts are never mixed. Without a power 1RM, the % of max is computed from the full lift's 1RM. Written programmes understand them too ("3x2 power clean @ 70%").

For the **jerk** and the **clean & jerk**, the athlete picks the overhead movement before the analysis; the choice is remembered:

Movement	What is measured
Split jerk	dip, drive velocity, bar displacement, lockout position, drop under the bar; posture: front and back knee in the split
Push jerk	the same, with the drop under the bar; posture: knee, hip and trunk at the catch
Squat jerk	the same, a larger drop under the bar; the end of the lift waits for the athlete to stand up; posture: angles at the bottom and depth (hip below the knee or not)
Push press	dip and drive, no dive under the bar: no drive-velocity goal and no drop under the bar; posture: trunk and elbows at lockout
Strict press	no dip: the app looks for the start of the arm press instead; velocity, bar displacement and lockout position; posture: trunk lean and elbows at lockout

Drop under the bar (jerk) is the distance from the top of the drive to the lowest point of the catch: almost zero in a push press, a few centimetres in a split jerk, much more in a squat jerk. The score only counts what applies to the chosen movement.



Movement list with power lifts and the overhead-movement choice

12 Strength lifts: squat, deadlift, bench press

LiftMetrics also tracks five strength lifts: **back squat, front squat, overhead squat, deadlift and bench press**. They are picked from the same “Movement” list as the Olympic lifts, with no extra menu, and are available everywhere: athlete 1RMs, session programmes (“5x5 back squat @ 80%”), history filters.

12.1 One video for the whole set

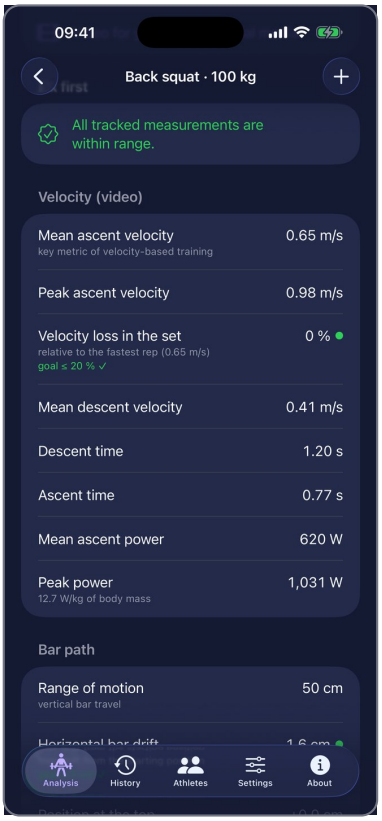
Unlike Olympic lifts (one video per lift), you film **the whole set in a single video**. The app detects each rep and analyses it separately: a set of 5 squats gives **5 analyses**, saved in one go and grouped in the history under the **SET** label, like a complex. Each rep keeps its own trimmed clip.

Movement	How a rep is detected
Squats, bench press	descent (at least 15 cm for the squats, 10 cm for the bench), bottom position, ascent until the bar stops at the top; the overhead squat starts with the bar overhead, arms locked
Deadlift	lift-off from the floor, rise of at least 25 cm, lockout, return to the floor

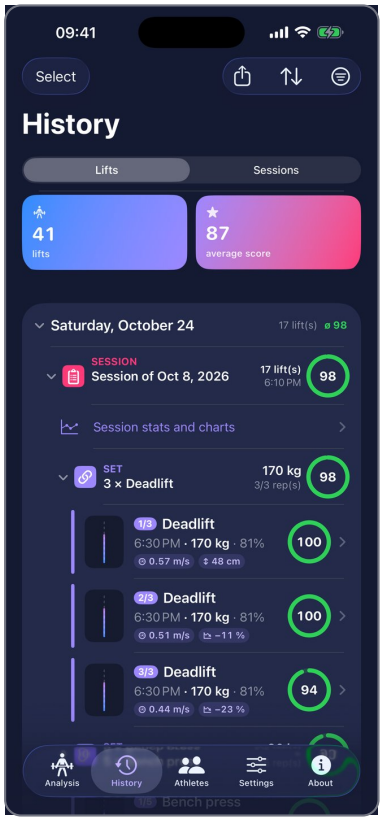
Phases are coloured as for Olympic lifts: **descent**, **ascent**, **sticking point** (if velocity dips during the ascent), **top**.



Set of 5 squats: velocity of each rep



Measurements for one rep



Set grouped in the history

Screenshots taken with a synthetic test video (5 squats, each slower than the last).

12.2 Measurements for each rep

Measurement	Definition	Hardware
Mean ascent velocity	vertical displacement / duration of the ascent: the reference measure of velocity-based training	iPhone
Peak ascent velocity	velocity peak during the ascent	iPhone
Velocity loss in the set	drop from the fastest rep (in %)	iPhone
Descent velocity and duration, ascent duration	tempo of the rep	iPhone
Sticking point	velocity and height (% of range) of the velocity dip during the ascent	iPhone
Range of motion	vertical travel of the bar	iPhone
Horizontal deviation	maximum bar deviation from its starting position	iPhone
Top position	final position relative to the start	iPhone
Mean and peak power	from the entered load	iPhone (+ sensor)
Bar tilt and rotation	one side higher, or ahead of the other	M5Stick

12.3 Score and goals

Criterion	Points	Default goal
Vertical bar path (horizontal deviation)	45	$\leq$ 5 cm squat, $\leq$ 4 cm deadlift, $\leq$ 10 cm bench press
Velocity loss in the set	35	$\leq$ 20 %
Bar tilt (sensor)	20	$\leq 3°$

On the bench press the bar naturally follows a J-curve, so the horizontal-deviation goal is wider. Velocities and range of motion depend on the load: they are not tracked by default but can be (for example a target mean-velocity zone for power work).

Why velocity loss?

In velocity-based training, the drop in mean velocity across a set reflects fatigue. Stopping the set once the loss exceeds a threshold (often 20 %) lets you dose the effort without going to failure. The app shows this loss rep by rep, in orange beyond the goal.

Filming a set

Side view, plate facing the camera fully visible, as for Olympic lifts. For squat and deadlift, lens at hip height; for bench press, camera at bench height. Start before unracking the bar (or before the first lift-off) and stop after the last rep. Comparison with reference athletes is not yet available for these lifts.

13 Reports

13.1 For each lift

Technique score out of 100. It summarises the bar path against the goals you set:

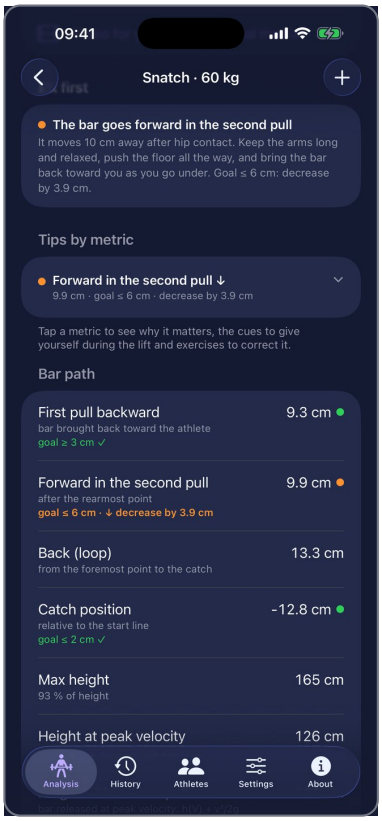
Criterion	Points
Forward in the second pull	30
Catch position	20
First pull backward	15
Drop under the bar	15
Bar tilt (sensor)	10
Bar path type	10
Clean & jerk: the jerk counts for 30 % of the score	

Each tracked measurement is **green** inside the range, **orange** slightly outside, **red** beyond. The app tells you whether to increase or decrease it, and by how much.

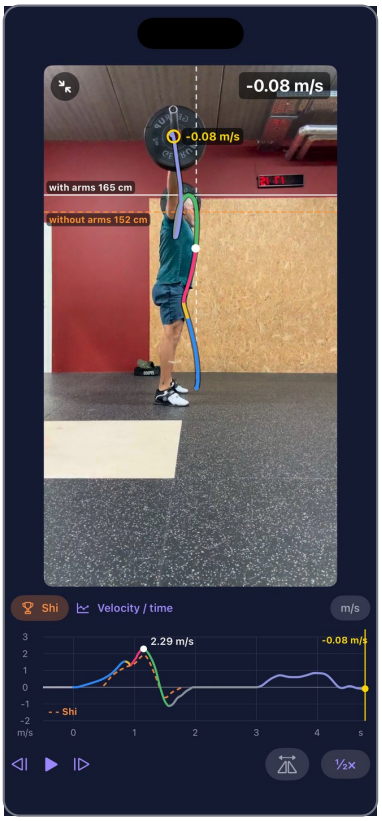
Coaching. The top three priorities, then for each measurement outside its goal: why it matters, cues to give during the lift and drills to fix it.

Replay. The video replays with the bar path drawing itself and the velocity shown at every instant; scrub through the lift by touching the curve. Slow motion ½ and ¼, frame by frame, mirror. The double arrow on the video switches to **full screen**.

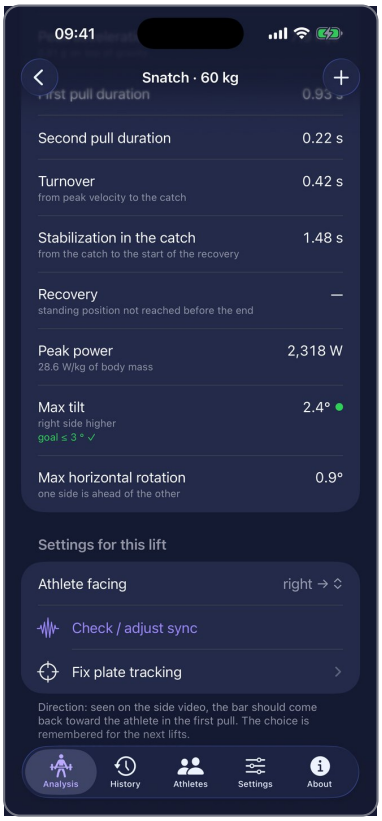
Charts. Curves against time: vertical velocity, acceleration, horizontal velocity, horizontal displacement, power (when the load is entered) and bar tilt (with the sensor). Velocity can be shown in m/s or as % of peak velocity.



Priorities and measurements



Full-screen replay



Velocity and timing measurements

13.2 History, sessions and tracking

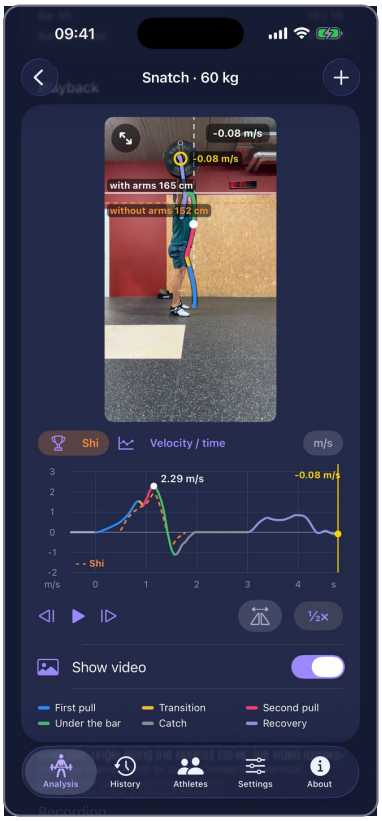
- **History**: every lift, grouped by day, with a bar-path thumbnail, the score and the key measurements; lifts of a complex are grouped and labelled. Sorting, filters (movement, athlete, make/miss), selection of lifts to analyse a session.
- **Sessions**: session statistics, average score, table of lifts and comparison of measurements.
- **Personal goals**: “From my lifts” turns the best quarter of made lifts into progression goals.
- **Overlaid bar paths**: all the session’s lifts of one movement drawn in the same frame, one figure per movement.
- **Review by movement**: recurring faults and trends (section 15).
- **Favourites and tags**, with sorting and filters (section 16).
- **History exports** to PDF (table of lifts).



History (grouped complex)



Session



Lift compared with Shi Zhiyong

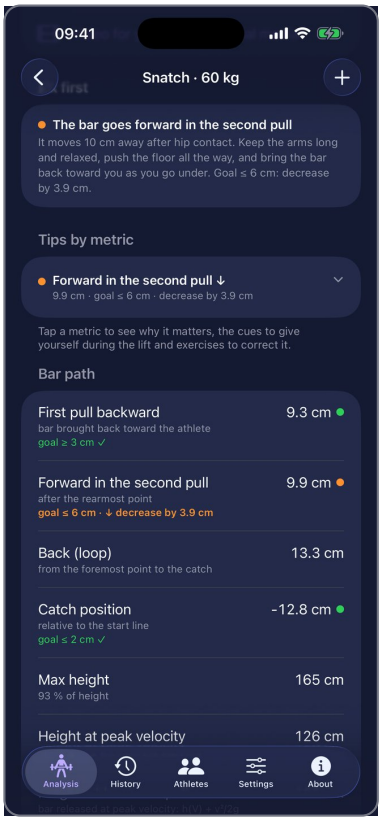
14 Coaching: tips and advice

Every tracked measurement has a goal range. When a lift falls outside it, **LiftMetrics** explains the fault and how to fix it. The default ranges come from elite lifters at maximal loads; each athlete can change them, or build them from their own best lifts.

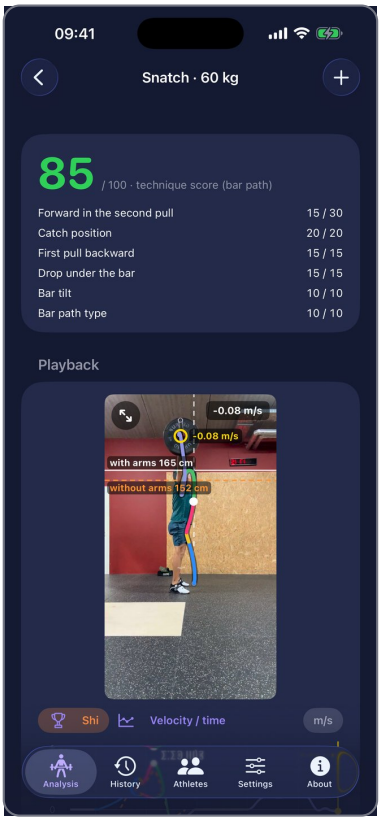
- **Green**: inside the goal, no tip. **Orange**: slightly outside (within the tolerance). **Red**: clearly outside.
- **Top 3 priorities**, red before orange, each in one sentence with the target and how much to change.
- **Tips by measurement**: for each fault, *why it matters*, *cues* to say during the lift and *drills* to fix it. “Too much” and “not enough” get different tips.

Fault	Triggered when	Typical cue and drill
Snatch and clean (power versions included)		
Bar not brought back in the 1st pull	back pull < 3 cm	lats tight, bar brushing the shins; pause pulls below the knee
Too much back pull	> 10 cm	lift off almost vertically; segmented pulls
Bar swings forward in the 2nd pull	> 6 cm (snatch), > 5 cm (clean)	long, relaxed arms, extend up not forward; hang snatch/clean, high pulls
Catch in front of the start line	> 2 cm in front	jump up, feet back in place; lifts with feet on a line
Catch behind the start line	> 10 cm behind	jump up, active arms over mid-foot; pause overhead/front squats
Big drop under the bar	> 16 cm (snatch), > 30 cm (clean)	pull yourself under, fast feet; tall snatch/clean, pull-unders
Bar tilted (sensor)	$> 3°$	symmetric grip and leg drive; unilateral leg work
High catch (posture)	full lift caught with the knee above 100°	ride the bar down into the full squat; pause squats, low catches only
Jerk		
Bar moves during dip and drive	> 3 cm	upright dip, weight on the heels, drive straight up; dip pauses, push press
Lockout off-centre	> 4 cm	head through, bar over the back of the neck; jerk balance, split holds
Dip too short / too deep	outside 14–22 cm	quarter-squat dip, immediate rebound; dip-and-drive, push press
Slow drive	< 1.4 m/s	drive hard to full ankle extension; explosive push press, power jerk
Squats, deadlift, bench press		
Bar drifts off vertical	> 5 / 4 / 10 cm	weight over the whole foot; pause and tempo work, goblet squat
Too much velocity loss in the set	> 20 %	every rep at full speed, stop the set earlier; shorter sets, longer rest

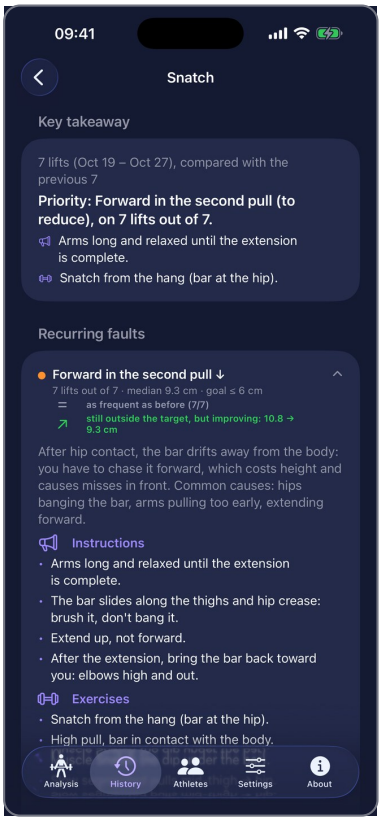
A measurement can also be tracked with a personal goal only (peak velocity, first-pull speed, arm contribution…): its tips then appear as soon as the athlete sets the goal.



Priorities with their targets



Score and breakdown



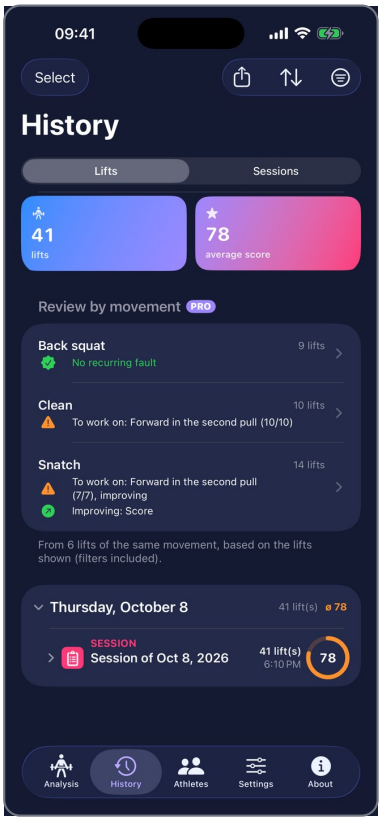
Tips for a recurring fault

15 Trend analysis: review by movement

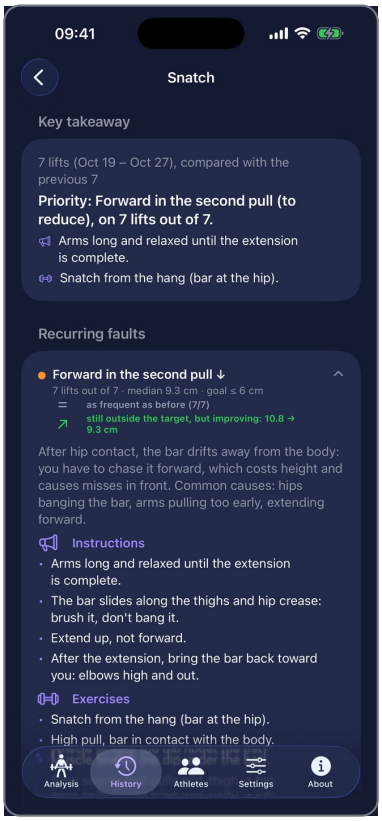
One lift shows a fault; ten lifts show a habit. At the top of the history, **LiftMetrics** Pro sums up each movement as soon as it has **6 lifts or more** (one line per athlete when there are several):

- **Recurring faults**: the measurements outside their goal on at least 30 % of the last 10 lifts, most frequent first, with the median value and the goal. Each fault says whether it is **becoming rarer, as frequent or more frequent** than in the 10 lifts before, or whether it is **new**. When the fault is still there but the value gets closer to the goal, it says so: “still outside the goal, but improving: 18.7 → 15.9 cm”.
- **Improving** and **getting worse**: the measurements whose distance to the goal changed clearly, the score, and the peak velocity **compared at equal load** (the load–velocity trend is removed, so heavier days do not look like a loss of speed).
- **Key takeaway**: the priority to work on, with one cue and one drill chosen for that recurring fault; every fault opens on its full tip (why, cues, drills).
- **Session PDF**: a “Review” part for each movement with at least 6 lifts in the session, compared with the athlete’s previous lifts.

The review follows the history filters: one athlete, one period, made lifts only…



Review by movement in the history

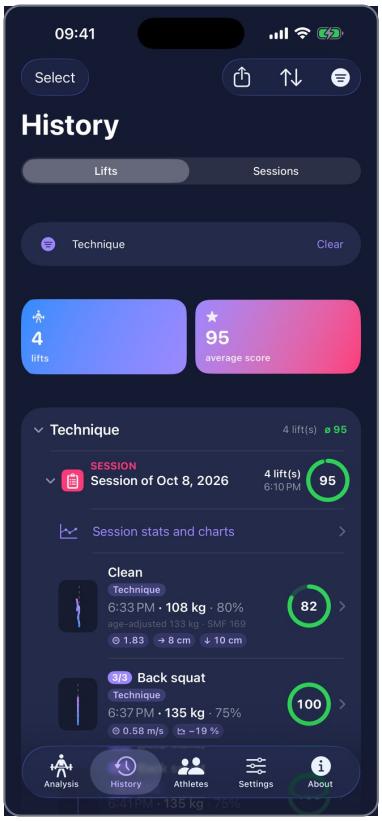


Priority, recurring fault and its trend

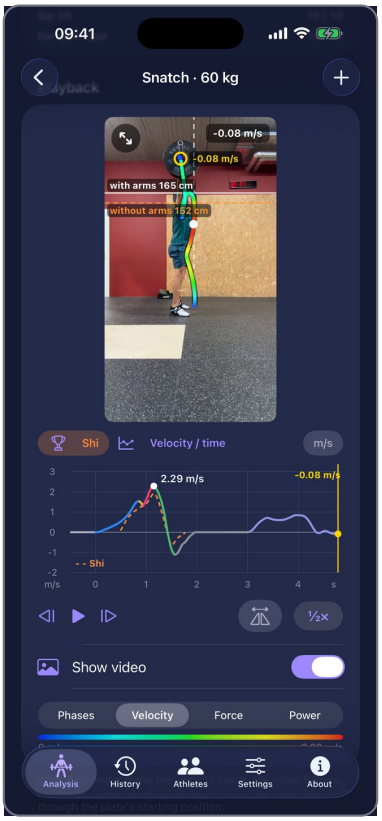
16 Tags, favourites and bar path colours

Favourites and tags. Any lift, complex, set or session can be starred and tagged (“competition”, “technique”, “heavy”…): swipe the row, long-press it, or use the star in the lift’s detail. The history can then be **sorted by tag** or **favourites first**, and **filtered** on favourites or on one tag — the review by movement follows.

Bar path colour. Under the replay, the path can be coloured by **phase** (default), or (Pro) by **velocity**, **force** or **power**, from blue (low) to red (high), with its scale. It shows at a glance where the bar accelerates, where it brakes and where the athlete pushes hardest. The picture, the video and the PDF follow the same choice.



History filtered on a tag



Replay coloured by velocity



Social-media picture, colour by velocity

17 Comparison with elite lifters

The app includes **reference snatches** digitised from scientific publications:

- **World elite** — 2018 IWF World Championships (Ashgabat): the best lifter in each category, **10 women and 9 men** (Om Yun Chol, Shi Zhiyong, Sohrab Moradi, Lasha Talakhadze, Sukanya Srisurat, Tatiana Kashirina. . .), with velocity curve, horizontal velocity and bar path.
- **National level** — Singapore national team (3 men, 3 women, at 80–90 % of 1RM): bar paths and reference values, closer to a good club level.

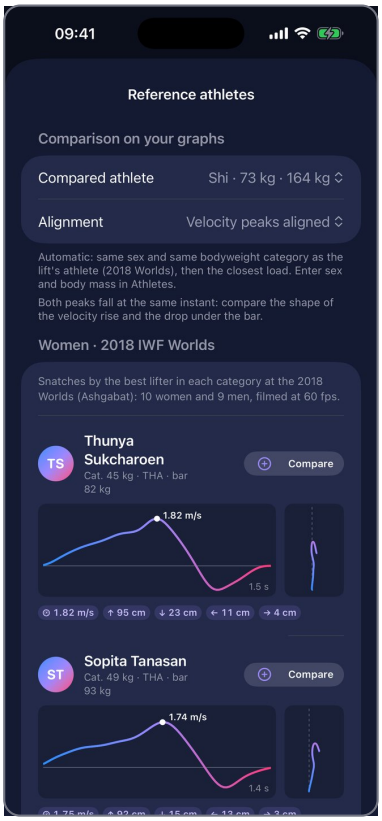
On the charts, the chosen athlete's velocity curve appears as a dashed orange line. The choice can be:

- **automatic**: same sex and bodyweight category as the athlete of the lift, then the closest load;
- **manual**: any athlete in the list.

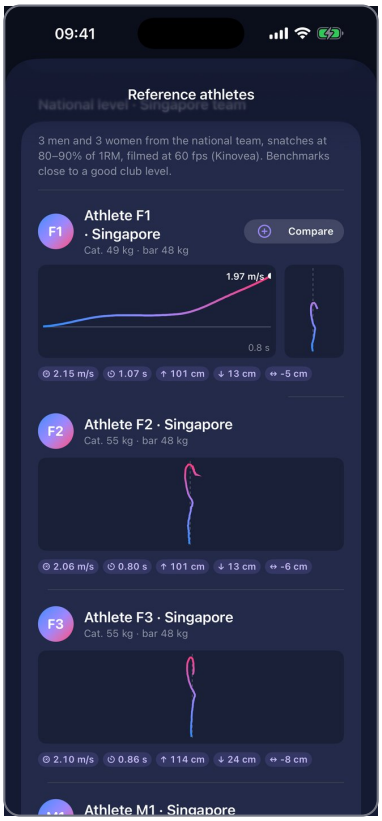
Time alignment, three options:

Peaks aligned	Both velocity peaks happen at the same instant: compare the shape of the acceleration and the turnover.
Lift-off and peak aligned	The reference curve is stretched or compressed: compare phases regardless of pull duration.
Lift-offs aligned	Real time: see who reaches peak velocity faster.

Clean and jerk. No athlete-by-athlete curves have been published; the app gives numerical benchmarks: peak clean height $\approx$ 59.5 % of height (men) and 62–68 % (women), drop under the bar 26 to 42 cm; in the jerk, dip $\approx$ 18 cm and drive velocity 1.53 m/s among elite lifters.



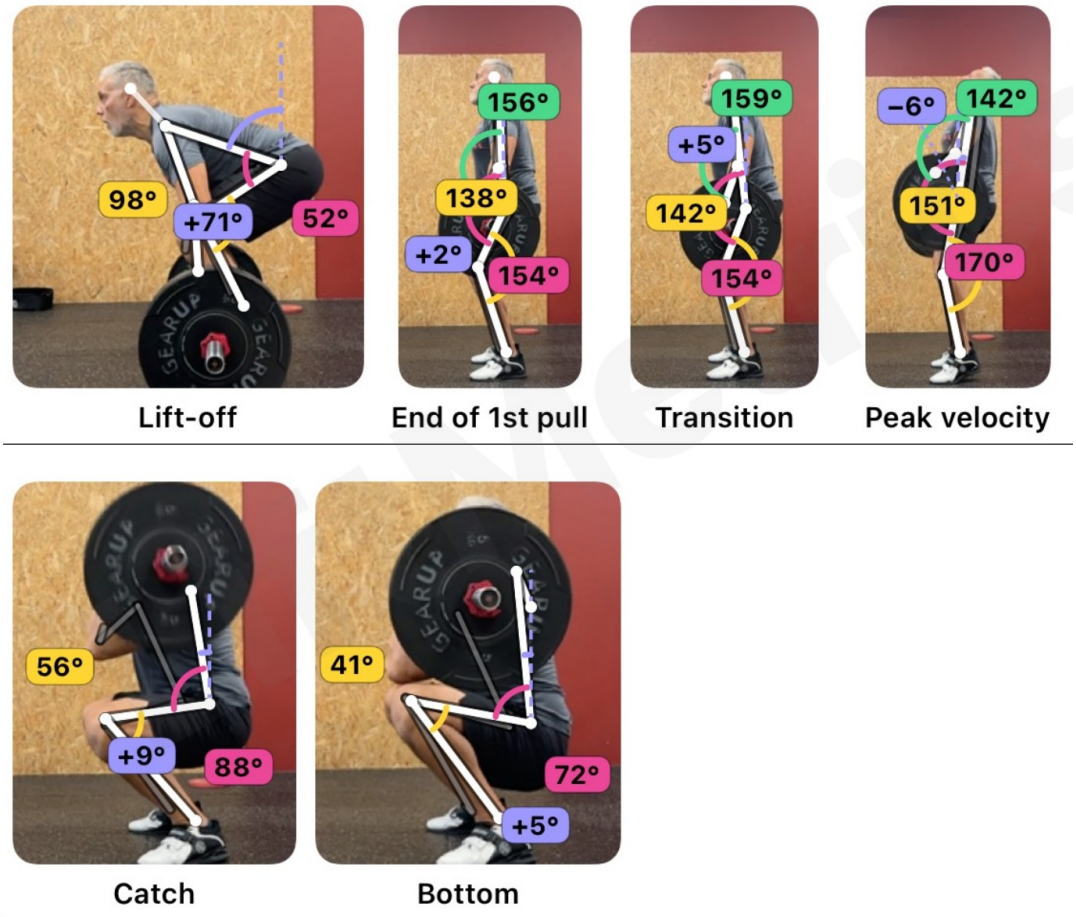
Reference athletes (Settings)



National level (Singapore)

18 Posture: body angles at key moments

Bar path tells *what* the bar does; posture shows *why*. On request, once the analysis is done, **LiftMetrics** Pro reads the athlete's body on the side-view video (Apple's on-device body detection) and measures the joint angles at the key moments it already knows from the bar: lift-off, end of the first pull, transition, peak velocity, catch and bottom position. It takes 10 to 20 extra seconds, and the result is saved with the lift.

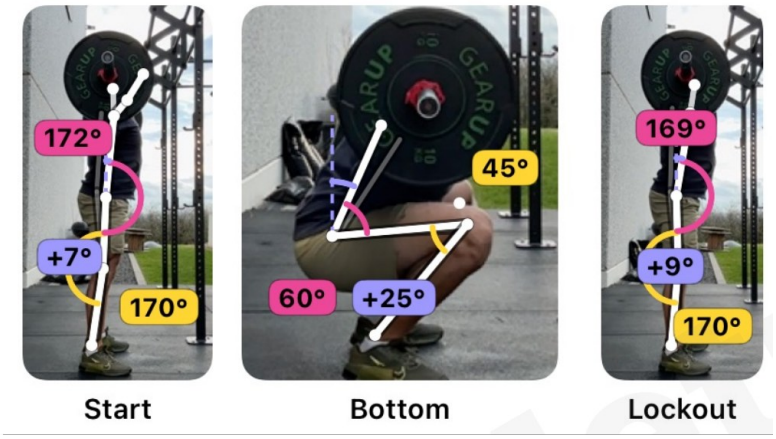


Clean with pause, 80 kg: one frame per key moment, skeleton and angles drawn on the video

Colour	Angle
Yellow	Knee (180° = straight)
Pink	Hip (180° = open)
Purple	Trunk relative to the vertical: + leaning forward, − leaning back
Green	Elbow, shown when it bends (arms pulling early)

- **Snatch, clean, clean & jerk**: lift-off, end of the first pull, transition, peak velocity, catch, bottom. In the example above, the hip reaches 170° at peak velocity while the elbows are already at 142°: the arms start pulling before the extension is complete.
- **Squats**: start, bottom, sticking point, lockout, and **depth** (hip below the knee or not).
- **Deadlift**: lift-off, bar at knee height, lockout.

- **Jerk**: front and back knee angles in the split. If the front leg is on the far side of the camera, the app says on which side to place the camera next time. Push jerk and squat jerk: angles at the catch, and depth for the squat jerk; push press and strict press: trunk lean and elbows at lockout.
- **Catch height**: the knee angle at the catch tells a power from a full lift. A full snatch or clean caught with the knee above 100° gets a “high catch” warning with its tip; a power caught below 80° is flagged as probably a full lift.



Back squat: start, bottom (hip below the knee), lockout

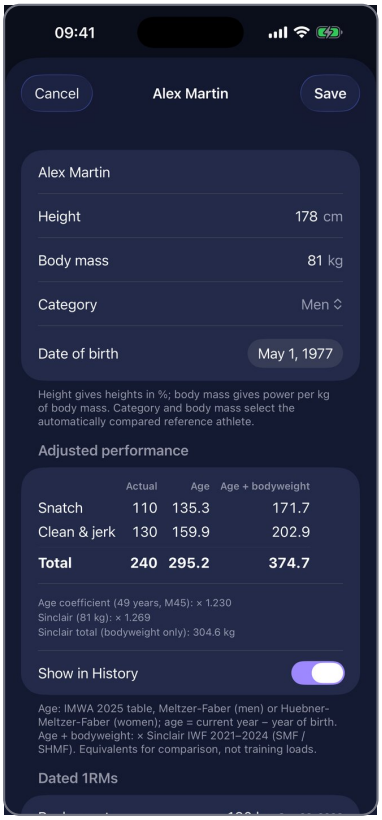
Angles are 2D measurements, side-on: accurate to a few degrees, less when the plate hides the legs or when the camera is not perpendicular to the bar — the app says so. They also appear in the lift’s PDF report.

19 Masters: age- and bodyweight-corrected loads

For snatch, clean and clean & jerk, **LiftMetrics** Pro shows three versions of a load, using the official Masters tables:

- **Actual** load;
- **Age-corrected**: multiplied by the IMWA 2025 age factor (Meltzer-Faber for men, Huebner-Meltzer-Faber for women, 30 to 95 years; Masters age = current year minus birth year);
- **Age + bodyweight**: also multiplied by the Sinclair coefficient (IWF, 2021–2024 cycle).

Example: a 49-year-old man (factor 1.230) weighing 81 kg (Sinclair 1.269) who snatches 80 kg → 98.4 kg age-corrected, 124.9 kg age + bodyweight.
The values appear on the athlete profile (with the M45, W50… category), under the load typed for a lift, complex or session, and as a discreet grey line in the history — flattering numbers, shown without taking over.



Athlete profile: adjusted performance

20 PDF report sample

Each lift exports to PDF. **Page 1**: header (movement, load, % of max, athlete, date), score and breakdown next to the bar path, priorities, then cues and drills. Then the full measurement table (and the posture frames when computed), and **all the curves on one page**, stacked on the same time axis (0 = lift-off) so the phases line up from one curve to the next. The glossary at the end only explains what the report contains. Sessions add a table of lifts and the overlaid bar paths, one figure per movement. Every page carries the LiftMetrics logo and a light watermark.





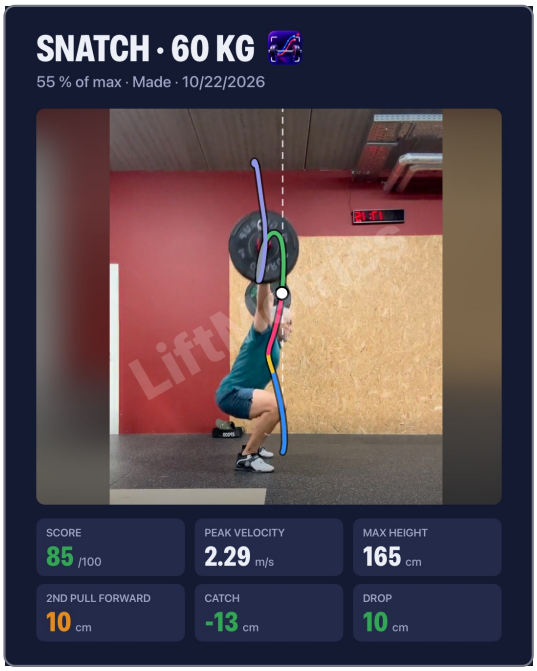
Page 1 and the curves page of the PDF report for a 60 kg snatch filmed with the M5Stick

21 Social media sample

Two ready-to-post formats:

- **Picture for your favourite social media** (1080×1350): photo of the catch with the bar path, score and 5 key measurements.
- **Video for your favourite social media** (post 1080×1350 or reel 1080×1920): the lift in slow motion, with the bar path drawing itself and the measurements appearing as they are reached.

Measurement colours follow the goals (green, orange, red). The LiftMetrics logo follows the title, and a light watermark crosses the photo or video. Sharing goes through the iPhone share sheet: Instagram, TikTok, WhatsApp…

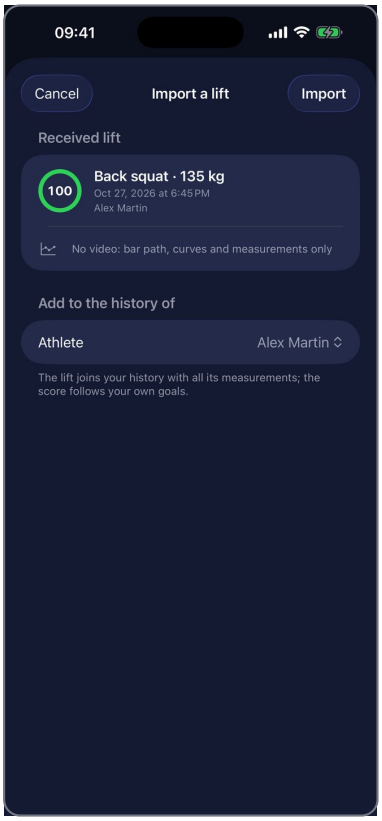


Content idea

A weekly “bar path of the week”: one heavy lift, its path and velocity, and the one cue that made the difference.

22 Sharing a lift

LiftMetrics file (.liftmetrics). An analysed lift can be sent via AirDrop, Messages, Mail or WhatsApp, with or without its trimmed clip. The recipient taps it: LiftMetrics opens, shows a preview and files the lift in their history, under the athlete of their choice. They get all the measurements, curves and replay; the score follows their own goals.
CSV. Curves (time, position, velocity) export for a spreadsheet or scientific processing.
Typical use: the athlete films at home or at the box, the coach receives the lift and reviews it with the same tools.



Importing a received lift

23 At competitions and events

LiftMetrics can support a competition or a training camp without disrupting it:

- **An analysis station beside the platform**: one iPhone on a tripod, in line with the bar, 3–5 m away. Lifts are filmed and analysed during the event; each athlete leaves with their report.
- **A warm-up station**: instant feedback between attempts (bar path, velocity, one thing to fix), useful for coaches.
- **Visuals for the event**: social-media pictures and videos of the best lifts, comparison with world references.
- **One file per athlete** (.liftmetrics or PDF) handed over after the event, which the athlete can open in their own app.
- **Strength events**: the same station works for squat, bench press and deadlift (velocity and bar path of every attempt).

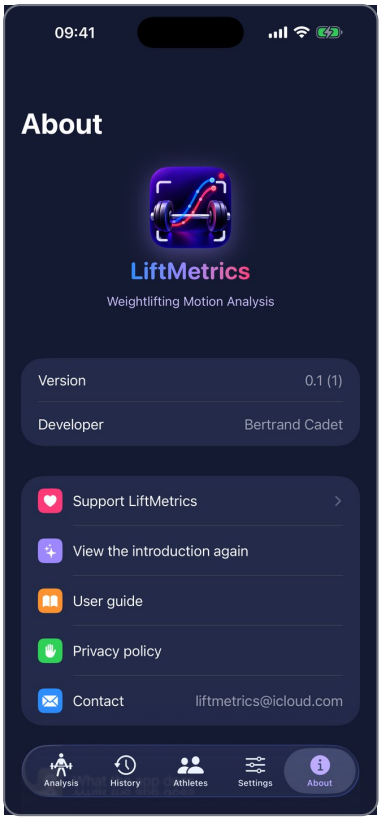
Practical points

- Free space beside the platform, in line with the bar, with nobody walking in front of the camera.
- Enough light on the platform; no backlighting behind the athlete.
- Athletes' consent to be filmed and for images to be shared.
- No data leaves the iPhone without a deliberate action: nothing is sent over the internet.

24 About

LiftMetrics is developed by **Bertrand Cadet**, a physics and computer science teacher and weightlifter. The app was born from a training need: to see and measure what the eye cannot catch at full speed, with equipment everyone already has in their pocket.
The app's **About** page brings together:

- Support LiftMetrics;
- View the introduction again;
- User guide and privacy policy;
- Contact: liftmetrics@icloud.com;
- What the app does and its limits;
- Data (everything stays on the device);
- Scientific sources.



About

Data and privacy

- No account, no ads, no tracking.
- Videos, measurements and profiles stay on the iPhone; they only leave it if the user exports or shares them.
- Full-length videos filmed by the app can be deleted automatically after analysis; a trimmed clip of the lift is kept.

Limits

Measurements are estimates: their accuracy depends on the camera angle (side view perpendicular to the bar), plate sharpness and how the sensor is mounted. Elite reference values are digitised from published figures and are therefore approximate. The app supports training and does not replace a coach's eye.

25 Pricing

Launch period	**Everything is free, Pro included.** Users who wish can support the project with a one-off in-app tip (3 amounts, e.g. €2.99, €4.99, €6.99). A discreet invitation appears after the 5th and 20th analysis.
Free, unlimited	Snatch, clean and clean & jerk analysis, power versions and all jerk variants: bar path, score, coaching, replay, curves; one lift at a time; history; one athlete profile; voice commands and automatic trim; social-media picture and video; receiving LiftMetrics files; M5Stick sensor.
LiftMetrics Pro (planned)	**One-time purchase, no subscription** (e.g. €9.99): squats, bench press and deadlift (VBT), comparison with elite lifters, complexes and programmed sessions, multiple athletes, personal goals, load-based charts and overlaid bar paths, review by movement (recurring faults and trends), bar path coloured by velocity, force or power, posture (body angles), Masters corrected loads, PDF reports, CSV export, sending LiftMetrics files.
Early users	Those who installed the app during the launch get LiftMetrics Pro for free.

Indicative prices, set at App Store release. Payments go through Apple (in-app purchases); the app has no access to any banking data.

26 Ambassador proposal

LiftMetrics is an independent project, built by a coach-developer, with no investors and no ads. The goal is to grow it through the people who actually lift. An ambassador would be the voice of the app in the CrossFit community and a direct partner in shaping it.

26.1 What LiftMetrics offers

- **LiftMetrics Pro for life**, and for the ambassador's coach and training partners.
- **Early access** to every new version through TestFlight, before public release.
- **Co-design**: direct line to the developer to build the features that matter for CrossFit athletes (for example barbell cycling, touch-and-go reps, complexes typical of CrossFit programming — the one-take complex, the overhead squat and the voice commands came from exactly this kind of need).
- **A sensor kit** (M5StickS3, 3D-printed sleeve mount, firmware pre-installed) shipped ready to use.
- **Visibility**: the ambassador featured on the app's Instagram account and in the app's communication, with her approval of every post.
- **A personal promo code or revenue share** on Pro purchases once paid features are switched on — terms to be agreed together.

26.2 What we would ask

- Use the app in training and give honest feedback, especially during the beta.
- Occasionally share lifts analysed with LiftMetrics (bar path, velocity) on social media, at a pace that suits her.
- Allow LiftMetrics to repost that content, credited.

Next steps

1. A short call or message exchange to discuss the idea and her expectations.
2. Invitation to the TestFlight beta (iPhone required).
3. Agreement on the terms above, adjusted to suit both sides.

Contact: Bertrand Cadet — liftmetrics@icloud.com — Instagram @lift_metrics

27 Scientific references

Reference values, default goals and comparison athletes are based on the following work.

Curves and bar paths of reference athletes

[1] Japan Weightlifting Association, Japan Institute of Sports Sciences (2018). *Kinematic characteristics of the snatch techniques of the top weightlifters — 2018 IWF World Weightlifting Championships (Ashgabat)*. Scientific Report.

[2] Ang C.L., Kong P.W. (2023). Field-Based Biomechanical Assessment of the Snatch in Olympic Weightlifting Using Wearable In-Shoe Sensors and Videos — A Preliminary Report. *Sensors*, 23(3), 1171. doi:10.3390/s23031171 (CC BY 4.0).

Clean and jerk benchmarks

[3] Liu G., He J., Ye J. et al. (2024). Comparative Analysis of the Kinematic Characteristics of Lunge-Style and Squat-Style Jerk Techniques in Elite Weightlifters. *Life*, 14(9), 1086 (CC BY).

[4] Ai Q., Bi Z., Liu G. (2018). Bar Heights Needed for Successful Lifts in Men's Weightlifters. *ISBS Proceedings*, 36, 217.

[5] Núñez Othón G. et al. (2015). Evaluación técnica del clin en halterofilia femenil. *Biotecnia*, 17(E1), 28–33.

[6] Prassas S., Fulton J. A Kinematic Analysis of the Jerk Technique in Olympic Weightlifting. *ISBS Proceedings*.

Masters and bodyweight corrections

[7] International Masters Weightlifting Association (2025). *IMWA Age Factors 2025*: Meltzer-Faber (men) and Huebner-Meltzer-Faber (women) coefficients, 30 to 95 years; *IMWA Handbook 2025–2026*.

[8] Meltzer D.E. (1994). Age dependence of Olympic weightlifting ability. *Medicine & Science in Sports & Exercise*, 26(8), 1053–1067.

[9] Huebner M., Meltzer D.E., Perperoglou A. (2019). Age-associated performance decline and sex differences in Olympic weightlifting. *Medicine & Science in Sports & Exercise*, 51(11), 2302–2308.

[10] Sinclair R.G. (1985). Normalizing the performances of athletes in Olympic weightlifting. *Canadian Journal of Applied Sport Sciences*, 10(2), 94–98; IWF Sinclair coefficients, 2021–2024 Olympic cycle.

Snatch reference values (default goals)

[11] Gourgoulis V. et al. (2000). Three-dimensional kinematic analysis of the snatch of elite Greek weightlifters. *Journal of Sports Sciences*, 18, 643–652.

[12] Kuhn M., Carlson K., Hoover D., Zebas C. Horizontal bar displacements of women weightlifters during the snatch. *ISBS Proceedings*.

[13] Wang C.-H., Chiu H.-T. (2008). Variability of barbell trajectory during snatch for an elite young female weightlifter. *ISBS 2008*, 644–647.

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A Glossary

Snatch	The bar goes from the floor to locked arms overhead in one movement.
Clean	The bar goes from the floor to the shoulders.
Clean & jerk	Clean then jerk: the bar goes from the shoulders to locked arms overhead.
Complex	Several movements chained within the same set (e.g. 1 snatch + 1 clean & jerk).
Overhead squat	Squat with the bar held overhead, arms locked (snatch grip).
Masters	Weightlifters aged 35 and over, competing in 5-year age categories (M35, W40…).
First / second pull	Phases of the bar's rise, before and after the knees re-bend under the bar.
Drop under the bar	Difference between the bar's peak height and its catch height.
Bar path type	Classic classification of the bar path depending on whether it crosses the starting vertical (types 1 to 3).
1RM	Maximum load lifted for one repetition.
Mean ascent velocity	Vertical displacement of the bar divided by the duration of the ascent (mean concentric velocity, MCV).
Velocity loss	Drop in mean ascent velocity compared with the fastest rep of the set.
Sticking point	Moment of the ascent where the bar slows down before accelerating again.
VBT	Velocity-based training: using bar speed to choose loads and manage fatigue.

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